Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Simplification of algebraic expressions: order of operations, brackets and word problems. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Second Term
Week	9
Topic	Simplification of Algebraic Terms
Subtopic	Simplification of algebraic expressions: order of operations, brackets and word problems
Theme	—
Curriculum objectives	• carry out order of operation in algebra • insert/remove bracket and simplify expression • solve word problems involving algebraic expression
Previous knowledge	• Addition and subtraction of algebraic terms • Multiplication and division of algebraic terms
Lesson sequence	• carry out order of operation in algebra — the teacher solves some questions in order of operation on the board; the student listen, contribute, and where necessary ask questions where they do not understand; the student copy worked examples into their notes; the student solve some questions independently; the teacher moves round to monitor their work, answer their questions, mark their notes and mark correction. Set for the students: 1. $4n + 3n \times 10$ 2. $32 + 8x \div 2$ 3. $3 \times 5x + 4x \div 2$ 4. $36y \div 6 + y + 2 \times 4y$ • insert/remove bracket and simplify expression — the teacher solves some questions on insertion, removal of bracket on the board; the student listen, ask questions on their areas of difficulty; the student copy worked examples into their notes; the student solve some questions independently; the teacher moves round to monitor their work, answer their questions, marks their notes and makes corrections. Set for the students: (a) $(12a - 7) + (4 - 5a)$ (b) $(8x - 2y) - (9x - 4y)$ • solve word problems involving algebraic expression — the teacher solves some questions on word problems involving algebraic expression on the board; the student listen, ask questions where they do not understand; the student solve some questions independently; the teacher moves round to monitor their work, answer their questions, mark their notes and mark correction. Set for the students: 1. A pen cost h naira and a crayon cost ₦20 less than a pen. (a) Find the cost of a crayon (b) Find the cost of a crayon and a pen 2. Envelopes cost d naira per pack, 4 note pad cost 20 naira more than a pack of envelopes. Find the total cost of a pack of envelope and notepad.
Instructional materials	• Chart showing algebraic expression.
Evaluation	• Simplify (a) $14pq \times r$ (b) $12x^2y \div 3x$ • Simplify $7 \times 3a - (3a + 5a) \times 2$

	• Simplify $12-(7-2f)-(3-f)$ • Simplify $(8a-5)+(14-3a)$ • A pen and a pencil altogether cost x naira. If a pen cost y naira, find the cost of 3 pencils.

Lesson objectives

1.	carry out order of operation in algebra
2.	insert/remove bracket and simplify expression
3.	solve word problems involving algebraic expression

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Simplification of algebraic expressions: order of operations, brackets and word problems

Class: JSS1 **Subject:** Mathematics **Week:** 9

Section 1: The Right Order for Algebraic Expressions

When you simplify an algebraic expression, you must deal with the operations in a fixed order. This order is often remembered by the word BODMAS:

- Brackets first
- Order (powers such as squares and cubes)
- Division and Multiplication (work from left to right)
- Addition and Subtraction (work from left to right)

The key idea is that you never just work from left to right. Brackets always come first, then powers, then multiply or divide, and only at the end do you add or subtract.

Multiplication and division are treated as equal in rank, so when you meet them together you simply work left to right. Addition and subtraction are also equal in rank, so when you meet them together you work left to right too.

Here is how the same steps work when the numbers are replaced by letters. For example, in the expression 3 + 2x, the multiplication 2x is done before the addition, because multiplication comes before addition. So 3 + 2x already means "2 times x, then add 3" — never "3 + 2, then times x". The letter x stands for a number, and the ordinary order of operations still applies to it.

A useful habit is to rewrite each term clearly. A term like 4y ÷ 2 means "4 times y, divided by 2", so it simplifies to 2y because 4 ÷ 2 = 2, then the y is carried along. Always work out the numbers first and keep the letter with its term.

Worked example 1 — multiply and divide, then add

Simplify 6x ÷ 2 + 4x.

Step 1: Deal with the division first, because division comes before addition.
6x ÷ 2 = 3x

Step 2: Now add the terms.
3x + 4x = 7x

So 6x ÷ 2 + 4x = 7x.

Worked example 2 — brackets come first

Simplify 3(x + 2y) − y.

Step 1: Deal with the bracket first. Multiply everything inside the bracket by the 3 in front.
3(x + 2y) = 3x + 6y

Step 2: Now subtract the y that is outside the bracket.
3x + 6y − y = 3x + 5y

So 3(x + 2y) − y = 3x + 5y.

Worked example 3 — powers come before multiply

Simplify 2x × x. Here the expression has no brackets, but remember that x × x is x^2 (a power). Multiplication of the letters gives a power.

Step 1: Multiply the letters.
2x × x = $2x^2$

So 2x × x = $2x^2$.

Practice

1. Simplify 4a × 3 + 5a.

2. Simplify $2(3p + q) - 2q$.
3. Simplify $8y \div 2 + y$.
4. Simplify $5m + 2 + 2m$, making sure to collect like terms in the right order.

Section 2: Brackets and Simplifying

Brackets are a way of grouping parts of an expression so we know what to deal with together. In algebra they do exactly the same job as in arithmetic: everything inside the brackets is treated as one group.

When a letter or number sits right in front of a bracket with no sign between them, it means multiply. For example, $3(x + 2)$ means 3 lots of $(x + 2)$, which is the same as $3 \times x + 3 \times 2$.

To remove brackets, we multiply the term outside the bracket by every term inside it. This is sometimes called expanding the bracket. Once the brackets are gone, we often collect the like terms (the terms with the same letter) to finish the simplification.

Watch a full example of removing brackets and then simplifying.

Example: Simplify $3(x + 2) + 4x$.

Step 1 — Remove the bracket by multiplying 3 by each term inside:
$3 \times x + 3 \times 2 = 3x + 6$

Step 2 — Bring down the rest of the expression:
$3x + 6 + 4x$

Step 3 — Collect like terms (the x terms together, the number terms together):
$3x + 4x + 6 = 7x + 6$

So $3(x + 2) + 4x = 7x + 6$.

Worked Example 1 — Removing two brackets

Simplify $2(x + 3) - 3(x - 1)$.

Step 1 — Remove the first bracket:
$2(x + 3) = 2x + 6$

Step 2 — Remove the second bracket. The minus sign in front means we subtract $3 \times (x - 1)$, so the -1 inside becomes $-3 \times -1 = +3$:
$-3(x - 1) = -3x + 3$

Step 3 — Put it together:
$2x + 6 - 3x + 3$

Step 4 — Collect like terms:
$2x - 3x + 6 + 3 = -x + 9$

So $2(x + 3) - 3(x - 1) = -x + 9$.

Inserting brackets (also called factorising) is the reverse of removing them. We look at every term and find the highest common factor — the largest number and the largest letter power that divides into all the terms.

For example, in $6x + 9$, both $6x$ and 9 divide by 3. So we write the 3 outside the brackets and put the remaining parts inside:

$6x + 9 = 3(2x + 3)$

Check: $3 \times 2x = 6x$ and $3 \times 3 = 9$, so it works.

Worked Example 2 — Inserting brackets

Factorise $8y + 12$.

Step 1 — Find the highest common factor of $8y$ and 12. The largest number that divides both 8 and 12 is 4.

Step 2 — Divide each term by 4:
$8y \div 4 = 2y$ and $12 \div 4 = 3$

Step 3 — Put the 4 outside the bracket:
8y + 12 = 4(2y + 3)

Step 4 — Check by expanding: 4 × 2y = 8y and 4 × 3 = 12. Correct.

Worked Example 3 — A bracket with a letter factor

Factorise 5a + 10a².

Both terms have a factor of 5 and also an a. The highest common factor is 5a.

5a ÷ 5a = 1 and 10a² ÷ 5a = 2a

So 5a + 10a² = 5a(1 + 2a).

Practice

1. Remove the brackets and simplify: 4(x + 5) + 2x
2. Remove the brackets and simplify: 5(y − 2) − 2(y + 1)
3. Insert brackets to factorise: 10x + 15
4. Insert brackets to factorise: 6m + 9m²
5. Remove the brackets and simplify: 2(3a + 4) + 3(a − 2)

Section 3: Order of Operations, Brackets, and Algebra in Words

When an algebraic expression contains several operations, you work through them in a fixed order. This order is the same one you use with numbers:

1. Do whatever is inside **brackets** first.
2. Then do **multiplication** and **division** from left to right.
3. Finally do **addition** and **subtraction** from left to right.

For example, in the expression $5 + 3 \times 2x$, there are no brackets, so you multiply before you add:
$5 + 3 \times 2x = 5 + 6x$.
The answer is $5 + 6x$ — you cannot add 5 and 6x because they are not like terms.

If we put brackets in, the order changes. In $(5 + 3) \times 2x$, you work out the bracket first:
$(5 + 3) \times 2x = 8 \times 2x = 16x$.

So brackets can change the meaning of an expression completely. Always clear them out first.

To **remove a bracket**, multiply each term inside the bracket by what sits in front of it. This is called expanding the bracket.

$3(2x + 4) = 3 \times 2x + 3 \times 4 = 6x + 12$

When there is a minus sign in front of the bracket, it multiplies every term inside, so every sign flips:
$-(3x - 5) = -3x + 5$

Here is a longer simplification that puts everything together:

[\n2(3x + 1) - (x - 4)\n]
Remove both brackets:

[\n= 6x + 2 - x + 4\n]
Collect the like terms (the x terms, then the numbers):

\[\n= (6x - x) + (2 + 4) = 5x + 6\n\]
Always simplify fully so no like terms are left over.

Algebra is useful for solving **word problems**. To turn words into an algebraic expression, follow these steps:

1. Choose a letter to stand for the unknown amount.
2. Translate each phrase into a piece of algebra:
   - "three times a number" means $3x$
   - "five more than a number" means $x + 5$
   - "a number decreased by two" means $x - 2$
   - "the sum of a number and four" means $x + 4$
3. Combine the pieces using the order of operations and brackets.

Here is an example. "A number is multiplied by 3, then 7 is added to the result." The number is x, so:

\[\n\text{result} = 3x + 7\n\]

Brackets matter in word problems too. "The sum of a number and 4, all multiplied by 2" is written as $2(x+4)$, not $2x+4$. The words "the sum of... all multiplied by" tell you to add first, so the addition goes inside the bracket.

Worked example 1 — Order of operations

Simplify $6+4\times 3a$.

There are no brackets, so multiply before adding:

$4\times 3a = 12a$.

Then $6+12a$.

6 and $12a$ are not like terms, so the answer is $6+12a$.

Worked example 2 — Removing brackets

Simplify $5(2x-3)-2(x+4)$.

Expand the first bracket: $5(2x-3) = 10x-15$.

Expand the second: $-2(x+4) = -2x-8$.

Put them together:

[\n10x - 15 - 2x - 8\n]

Collect like terms:

[\n(10x - 2x) + (-15 - 8) = 8x - 23\n]

Answer: $8x-23$.

Worked example 3 — Word problem

"Twice the sum of a number and 5" — write this as an expression and simplify it.

Let the number be x.

"The sum of a number and 5" is $x+5$.

"Twice" means multiply by 2, so the whole sum goes inside a bracket:

[\n2(x + 5)\n]

Expand:

[\n2x + 10\n]

So the expression is $2x+10$.

Practice

1. Simplify: $9+2\times 4x$
2. Remove the brackets and simplify: $3(2x+1)+(x-5)$
3. A number x is increased by 3, and then the result is multiplied by 4. Write this as an algebraic expression and simplify it.
4. The difference between a number y and 2 is multiplied by 5. Write this as an algebraic expression and simplify it.

Check your work

Attempt every question before you read this page.

The Right Order for Algebraic Expressions

1. 17a

   Multiplication comes before addition, so do 4a × 3 first.
   4a × 3 = 12a
   Now add: 12a + 5a = 17a.
2. 6p

Brackets come first. Multiply everything inside the bracket by 2:
$2(3p + q) = 6p + 2q$
Now subtract the outside term: $6p + 2q - 2q = 6p$.

3. 5y

   Division comes before addition, so do $8y \div 2$ first.
   $8y \div 2 = 4y$
   Now add: $4y + y = 5y$.

4. 7m + 2

   Addition and subtraction have equal rank, so work from left to right.
   $5m + 2m = 7m$, then keep the $+2$.
   So the expression is $7m + 2$.

Brackets and Simplifying

1. 6x + 20

   Remove the bracket: $4(x + 5) = 4x + 20$. Bring down the rest: $4x + 20 + 2x$. Collect like terms: $4x + 2x + 20 = 6x + 20$.

2. 3y − 12

   Remove the first bracket: $5(y - 2) = 5y - 10$. Remove the second: $-2(y + 1) = -2y - 2$. Together: $5y - 10 - 2y - 2$. Collect like terms: $5y - 2y - 10 - 2 = 3y - 12$.

3. 5(2x + 3)

   The highest common factor of 10x and 15 is 5. Divide each term by 5: $10x \div 5 = 2x$ and $15 \div 5 = 3$. Write 5 outside the bracket: $5(2x + 3)$. Check: $5 \times 2x = 10x$ and $5 \times 3 = 15$.

4. 3m(2 + 3m)

   Both terms share a factor of 3 and of m, so the highest common factor is 3m. Divide: $6m \div 3m = 2$ and $9m^2 \div 3m = 3m$. Write: $6m + 9m^2 = 3m(2 + 3m)$. Check by expanding: $3m \times 2 = 6m$ and $3m \times 3m = 9m^2$.

5. 9a + 2

   Remove the first bracket: $2(3a + 4) = 6a + 8$. Remove the second: $3(a - 2) = 3a - 6$. Together: $6a + 8 + 3a - 6$. Collect like terms: $6a + 3a + 8 - 6 = 9a + 2$.

Order of Operations, Brackets, and Algebra in Words

1. $9 + 8x$

   Multiply before adding: $2 \times 4x = 8x$. Then $9 + 8x$. 9 and 8x are not like terms, so the answer is $9 + 8x$.

2. $7x - 2$

   Expand: $3(2x + 1) = 6x + 3$ and $(x - 5) = x - 5$. Together: $6x + 3 + x - 5$. Collect like terms:
   $(6x + x) + (3 - 5) = 7x - 2$.

3. $4(x + 3) = 4x + 12$

   "Increased by 3" gives $x + 3$. "Then multiplied by 4" means the whole sum goes inside a bracket: $4(x + 3)$. Expanding gives $4x + 12$.

4. $5(y - 2) = 5y - 10$

   "The difference between a number y and 2" means $y - 2$. "Multiplied by 5" means the whole difference goes inside a bracket: $5(y - 2)$. Expanding gives $5y - 10$.

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-3d48f87b	**Date**	
Subtopic	Simplification of algebraic expressions: order of operations, brackets and word problems	**Rater**	

The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____